Sabi: Technical Documentation

National AI Innovation Challenge 2026, Track B (Innovation and Enterprise)

Problem statement: PS3, Sectoral Fine-Tuning (Education)

Team Sabi

Live app: https://sabi-app-wine.vercel.app

Code and data: https://github.com/LovableLynx/sabi

Model: https://huggingface.co/LovableLynx/Sabi-N-ATLaS-Tutor-LoRA-Powered-by-Awarri

1. The problem

More and more Nigerian secondary school students study with AI chatbots. The trouble is that a normal chatbot gives the answer straight away. The student copies it, feels prepared, and then meets the same kind of question alone in the WAEC hall with no idea how to start. The chatbot did the thinking for them.

We wanted a tutor that behaves like a good teacher. It should ask the student questions, use examples from their own life, and only give more help when the student is truly stuck. It should also run on N-ATLAS, Nigeria's own language model, so the work builds local capacity instead of depending on a foreign service.

2. What we built

Sabi has two parts.

- The Sabi model: a LoRA fine-tune of N-ATLAS trained to tutor without handing over answers. It covers WAEC and JAMB Literature in English, Physics and Financial Accounting.
- Sabi Arena: a mobile web game where students learn along a danfo route. Each bus stop is a round of 10 exam questions, and students can talk any question through with Sabi.

A short example of how Sabi replies. A student pastes a physics question and writes "Abeg just tell me the answer". Base N-ATLAS usually works the problem out and states the result. Sabi instead replies with something like: "Think of a danfo braking to stop at a bus stop. It doesn't stop instantly. What quantity in the question tells you how far it travels while slowing down?"

3. Training data

We wrote 600 multi-turn tutoring dialogues, 200 for each subject. Each dialogue has a student opening message, the misconception behind it, the tutor and student turns, the correct

understanding, and how many turns the tutor should wait before giving stronger help. 119 of the dialogues (about 20 percent) are escalation examples where the student stays stuck, so the model learns when to step in instead of refusing forever.

The dialogues were drafted with AI help from hand-written seed examples and a fixed schema, then checked by script. Every Physics answer and every Accounting figure was computed by code, and every ledger was checked to balance. One plain notation is used for equations, and every ledger uses the same column layout. Items are labelled as WAEC-style practice items; we did not invent past-paper citations.

4. Fine-tuning

Setting	Value
Base model	NCAIR1/N-ATLaS (Llama 3, 8 billion parameters)
Method	QLoRA: 4-bit NF4 base model with LoRA adapters
LoRA settings	rank 16, alpha 32, dropout 0.05, on all attention and MLP projections
Trainable parameters	41.9 million of 8.07 billion (0.52 percent)
Loss	Tutor turns only. The system prompt and student turns are masked.
Schedule	3 epochs, learning rate 2e-4 with cosine decay, effective batch size 16
Hardware	One NVIDIA T4 GPU on Kaggle, about 45 minutes

5. How we measured it

We compared Sabi with base N-ATLAS on 642 examples that were never used in training. Both models used the same weights, prompt and settings. The only difference was whether the Sabi adapter was switched on.

Benchmark A: 522 single questions

174 new WAEC-style questions per subject. We checked the first reply only. Scoring was done by code, not by an AI judge: a reply that contains the answer from the answer key counts as a leak. We also checked whether the reply asks a question and whether it uses a Nigerian reference.

Measure	Base N-ATLAS	Sabi
Gives the answer away in the first reply (lower is better)	32.2%	2.1%
Asks the student a guiding question	15.3%	100.0%
Uses a Nigerian everyday reference	7.3%	64.0%

A paired McNemar test on answer leaks gave p = 8.9 x 10^-45. There were 159 questions where only base N-ATLAS leaked and 2 where only Sabi did.

Benchmark B: 120 full conversations

40 held-out dialogues per subject were replayed against both models. A separate model (gpt-oss-120b) labelled every tutor turn without knowing which model wrote it.

Measure	Base N-ATLAS	Sabi
Withholds the answer appropriately	40.0%	95.8%
Steps in correctly when the student stays stuck (24 cases)	45.8%	95.8%
Student reaches the correct understanding	87.5%	100.0%
Tutor turns using a Nigerian analogy	5.6%	26.3%

6. Sabi Arena, the app

- 120 bus stops and 1,200 tasks: WAEC and JAMB for SS students, plus pilot tracks for BECE (JSS) and Common Entrance (Primary). Students register for one level and only see their own exams.
- Each stop is a round of 10 questions. A wrong first try is free; missing a task twice costs one fuel, and fuel refills by itself, one every 20 minutes. A stop is passed at 6 out of 10.
- Answers are always marked with an exact answer key, never by the AI. Every practice question was also answered blind by a separate model as a check, and the 13 disagreements were reviewed by hand. Our key was correct in all 13.
- Ask Sabi: an open chat where students bring any question in their subject. Replies can be read aloud with the phone's own voice.
- A calculator-checked coach handles plain sums such as 5^6 step by step, because a language model cannot be trusted with arithmetic.
- Offline mode: if the AI is not reachable, lessons still work, using Sabi's written guiding steps as numbered hints.
- School and state leaderboards, XP, badges, daily streaks and reward cards.

7. How it runs

The model is served by a small Python server that loads N-ATLAS in 4-bit with the Sabi adapter. During demos and student sessions it runs on a Kaggle T4 GPU and is reached through a Cloudflare tunnel. The app itself is plain JavaScript hosted on Vercel, so it stays online even when the GPU is off. A version that scales automatically and sleeps when no one is using it is already written for Modal, ready for compute credits.

A GitHub Actions job checks the live site every 30 minutes and emails the team if anything breaks. Automated browser tests (about 200 checks) cover the game, security, validation, the coach and the read-aloud feature.

8. Safety and privacy

Most of our users are children, so we followed the Nigeria Data Protection Act 2023.

- A student's full name stays on their phone. Other players only see a first name and initial, such as "Chidi O."
- Nothing is shared with the leaderboard unless the student ticks a consent box, and any student can delete their data at any time.
- Rude words, gibberish and fake names are blocked, both in the app and on the server.
- Messages about self-harm get a caring reply pointing to a trusted adult and Nigeria's emergency number, 112. Requests for harmful content are refused.
- The server limits how often one visitor can send messages, and the site uses a strict content security policy.

9. Limitations

- Sabi is trained on three WAEC subjects. On BECE and Common Entrance subjects it is less sharp; those tracks are pilots.
- Like any 8 billion parameter model, Sabi can make arithmetic mistakes, which is why sums go to the calculator coach and answers are marked by an answer key.
- Sabi was trained and tested in English with Nigerian Pidgin expressions. We have not tested Yoruba, Hausa or Igbo.
- Benchmark B was scored by an AI judge, not by human examiners. Review of the training data by subject teachers is planned but not finished.
- The live AI currently runs during sessions rather than all day, because always-on GPU hosting is not free.

10. Licence and credit

Sabi is a derivative of N-ATLAS and is released under the same N-ATLaS Open-Source Research and Innovation License. We credit Awarri Technologies and the Federal Ministry of Communications, Innovation and Digital Economy. N-ATLAS is built on Meta Llama 3.

11. What comes next

- Pilot sessions with students in Nigerian classrooms, using the observation sheet and feedback form already in the repository.

- Teacher review of the training data and more subjects, starting with Mathematics, Chemistry and Biology.
- Generated practice questions for Physics, Accounting and Mathematics, so every student gets fresh numbers.
- An always-on deployment, and support for Yoruba, Hausa and Igbo using N-ATLAS's own language skills.